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PS Seminar 1

rimona. vojica @ fmi. unibz. ro

Evaluare :

50 %	Examen	
30 %	proiect	
20 %	activitate	10 % test (in materiale) 10 % lab ? bonus

alex marisarei. quarto. pub / mes - ps - fmi

R Studio

Probabilități

multimea stărilor
↓
 $(\Omega, \mathcal{F}, P)$

ex

1) Aruncarea unui zar o dată

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

2) Aruncarea unui zar de două ori

$$\Omega = \{(1, 1), (1, 2), \dots, (1, 6) \dots (6, 6)\}$$

...

n aruncări

$$3) \Omega = \{(w_1, w_2, \dots, w_n) \mid w_i \in \{1, 2, 3, 4, 5, 6\}\}$$

Un **eveniment** este o mulțime de stări
eveniment atomic (o rg. stare)

↗ σ -algebră

$$\mathcal{F} \subseteq \mathcal{P}(\Omega)$$

↑ mulțimea evenimentelor posibile

- 1) evenimente : $A = \{2, 4, 6\}$
 $B = \{2, 3, 5\}$
 $C = \{6\}$ store (element)
↑
eveniment atomic

- 2) even $A = \{(2, 4), (2, 6)\}$
 $B = \{(i, i) \mid i = \overline{1, 6}\}$

Def $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ s.n. σ -algebră dacă

- 1) $\emptyset, \Omega \in \mathcal{F}$ complementare A^c
- 2) $A \in \mathcal{F} \Rightarrow \bar{A} \in \mathcal{F}$ reuniune numărabilă
- 3) $A_1, A_2, \dots, A_n, \dots \in \mathcal{F}_n \Rightarrow \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{F}$
sau
 $((A_n)_n \subset \mathcal{F})$

exemplu

$$\Omega = \{a, b, c\}$$

$$\mathcal{P}(\Omega) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \Omega\}$$

Cea mai mare σ -algebră pe mulțimea Ω

$$\mathcal{F}_0 = \{ \emptyset, \Omega \}$$

$$\mathcal{F}_1 = \{ \emptyset, \Omega, \{a, b\}, \underbrace{\{c\}}_{\text{complementara lui } \{a, b\}} \}$$

$$\mathcal{F}_2 = \{ \emptyset, \Omega, \{a, b\}, \{c\}, \{b\}, \{a, c\}, \{b, c\}, \{a\} \}$$

Temă $\Omega = \{a, b, c, d\}$

Generati toate σ -algebrele

Bonus Algoritm in R pt a genera toate

σ -algebre cu 5 elemente + caz general (3 pt)

Temă

$$\Omega = \{a, b, c, d\}$$

Generati toate σ -algebrele
#

$$\mathcal{F}_1 = \{\emptyset, \Omega\}$$

$$\mathcal{F}_2 = \{\emptyset, \Omega, \{a\}, \{b, c, d\}\}$$

$$\mathcal{F}_3 = \{\emptyset, \Omega, \{b\}, \{a, c, d\}\}$$

$$\mathcal{F}_4 = \{\emptyset, \Omega, \{c\}, \{a, b, d\}\}$$

$$\mathcal{F}_5 = \{\emptyset, \Omega, \{d\}, \{a, b, c\}\}$$

$$\mathcal{F}_6 = \{\emptyset, \Omega, \{a, b\}, \{c, d\}\}$$

$$\mathcal{F}_7 = \{\emptyset, \Omega, \{a, c\}, \{b, d\}\}$$

$$\mathcal{F}_8 = \{\emptyset, \Omega, \{a, d\}, \{b, c\}\}$$

~~$$\mathcal{F}_9 = \{\emptyset, \Omega, \{b, c\}, \{a, d\}\}$$~~

~~$$\mathcal{F}_{10} = \{\emptyset, \Omega, \{b, d\}, \{a, c\}\}$$~~

anulati

$$F_9 = \{ \emptyset, \Omega, \{a, b\}, \{c\}, \{d\}, \\ \{a, b, c\}, \{a, b, d\}, \{c, d\} \}$$

$$F_{10} = \{ \emptyset, \Omega, \{a, c\}, \{b\}, \{d\}, \\ \{a, c, b\}, \{a, c, d\}, \{b, d\} \}$$

$$F_{11} = \{ \emptyset, \Omega, \{a, d\}, \{b\}, \{c\}, \\ \{a, b, d\}, \{a, d, c\}, \{b, c\} \}$$

$$F_{12} = \{ \emptyset, \Omega, \{b, c\}, \{a\}, \{d\}, \\ \{a, b, c\}, \{a, d, c\}, \{a, d\} \}$$

$$F_{13} = \{ \emptyset, \Omega, \{b, d\}, \{a\}, \{c\}, \\ \{a, b, d\}, \{c, b, d\}, \{a, c\} \}$$

$$F_{14} = \{ \emptyset, \Omega, \{c, d\}, \{a\}, \{b\}, \\ \{a, c, d\}, \{b, c, d\}, \{a, b\} \}$$

$$F_{15} = \{ \emptyset, \Omega, \{a\}, \{b\}, \{c\}, \{d\}, \\ \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \\ \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \\ \Omega \}$$